

Lab Report Questions

Instructions:

- Create your new repository on GitHub and push your lab solution to GitHub with new branch for each lab set.
- Share your repository to instructor.basanta@gmail.com as a collaborator.
- Print and attach your updated GitHub screenshot into lab set

Format

- Cover (Print)
- GitHub Screenshot
- Lab Question –hand written
- Introduction –hand written
- Syntax and Format –hand written
- Code –hand written
- Output(Print)- With full URL

Lab-1

1. Write a PHP program to create variable with different **datatypes**,
 - a. Print all the data using **echo** and **print**
 - b. Display content of array using **print_r** and **var_dump**
 - c. Display result of **checking data types**.
2. Calculate area of circle taking radius input and defining value of PI as **constant** value.
3. Write a function that takes an integer **minutes** and converts it to **seconds**.
4. Create a function that takes **two** numbers as arguments and returns their **sum**.
5. Write a function that takes the **base** and **height** of a triangle and return its **area**.
6. Create a function that takes the **age** in **years** and returns the **age** in **days**.
7. Write a PHP program to calculate the total points obtained by a football team based on the number of **wins, draws, and losses**. Create a user-defined function that accepts the number of wins, draws, and losses as parameters and calculates the total points according to the following rules:
 - **Win:** 3 points
 - **Draw:** 1 point
 - **Loss:** 0 points

Create an HTML form to accept the number of wins, draws, and losses from the user. On form submission, process the input using PHP, call the user-defined function, and display the **total number of games played and total points obtained by the team**. Ensure that appropriate input validation is applied to prevent invalid or negative values.

8. Create a function that takes two strings as arguments and return either true or false depending on whether the total number of characters in the first string is equal to the total number of characters in the second string.
9. Create a function that returns true if an integer is evenly divisible by 5, and false otherwise.
10. Write a function that returns the length of a string. **Make your function recursive.**
11. Write a function that accepts base (decimal), height (decimal) and shape ("triangle", "parallelogram") as input and calculates the area of that shape.
12. Create a function that takes an array and a string as arguments and returns the index of the string.
13. Given an index and an array, return the value of the array with the given index.
14. A typical car can hold **four passengers** and **one driver**, allowing **five people** to travel around. Given **n** number of people, return how many cars are needed to seat everyone comfortably?
15. Write a PHP program to compute the sum of the two given integer values. If the two values are the same, then returns triple their sum.
16. Write a PHP program to get the absolute difference between n and 51. If n is greater than 51 return triple the absolute difference.
17. Write a PHP program to create a new string where 'if' is added to the front of a given string. If the string already begins with 'if', return the string unchanged.

Sample Input:

"if else"
"else"
"if"

Sample Output:

if else
if else
if

18. Write a PHP program to create a new string which is 4 copies of the 2 front characters of a given string. If the given string length is less than 2 return the original string.

Sample Input:

"C Sharp"
"JS"
"a"

Sample Output:

C C C C
JSJSJSJS
a

19. Write a PHP program to create a new string with the last char added at the front and back of a given string of length 1 or more.

Sample Input:

"Red"
"Green"
"1"

Sample Output:

dRedd
nGreenn
111

20. Write a PHP program to create a new string taking the first 3 characters of a given string and return the string with the 3 characters added at both the front and back. If the given string length is less than 3, use whatever characters are there.

Sample Input:
"Python"
"JS"
"Code"
Sample Output:
PytPythonPyt
JSJSJS
CodCodeCod

21. Write a PHP program to check the largest number among three given integers.
22. Write a PHP program to convert the last 3 characters of a given string in upper case. If the length of the string has less than 3 then uppercase all the characters.

Sample Input:
Nepal
Npl
Bca
Bachelor
Sample Output:
NePAL
NPL
BCA
BacheLOR

23. Write PHP Script to display following array into HTML table.

```
$info = [ 'name' => 'Ram Bahadur', 'address' => 'Lalitpur', 'email' => 'info@ram.com',  
'phone' => 98454545, 'website' => 'www.ram.com'];
```

Name	Ram Bahadur
Address	Lalitpur
Email	info@ram.com
Phone	98454545
Website	www.ram.com

24. Write PHP Script to display student mark sheet into table, store data into PHP multidimensional array.

Mark Ledger

SN	Name	Roll	Web Tech II	DBMS	Economics	DSA	Account	Total	Result
1	Rajesh	25	56	89	57	64	98	364	pass
2	hari	5	56	89	57	64	98	364	pass
3	Shyam	6	54	79	57	69	98	357	pass
4	Rita	10	16	89	56	64	98	323	fail
5	Gita	4	56	89	57	69	98	369	pass
6	Sita	24	56	99	57	24	98	334	fail
7	Sita	24	56	99	57	24	98	334	fail
8	Sita	24	56	99	57	24	98	334	fail

Alternate color

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4	Rita	10	16	89	56	64	98	323	fail
5	Gita	4	56	89	57	69	98	369	pass
6	Sita	24	56	99	57	24	98	334	fail
7	Sita	24	56	99	57	24	98	334	fail
8	Sita	24	56	99	57	24	98	334	fail

25. Write PHP Script to create and validate login form with username and password.

26. Write PHP Script to create and login form to implement session and cookie.

27. Write PHP script to upload CV with following details:

- File type: PDF & DOCS
- File size less than 1 MB

28. Write PHP script to upload Profile Image with following details:

- File type: PNG & JPEG
- File size less than 500 KB

29. Write a PHP script to validate and register (store) user with following data.

- Username with minimum 8 character
- Valid email Address
- Validate Date of Birth
- Valid phone length

30. Write a PHP Script to validate, store, list, and update and delete record from database using MySQL database with following fields.

- id,name,rank,status,image,created_by,updated_by,created_at,updated_at

31. Write PHP program to perform CRUD operation for the following tables.

courses

id
title
duration
status
created_at
updated_at

students

id
name

course_id (use foreign key)
fee
rollno
phone
address
dob
status
created_at
updated_at

32. Write PHP Script to take user marks using form and generate mark sheet when user input all data. (Please look at last page for mark sheet format)
33. Write a server-side script to send email notification for specified email address.
34. Write PHP script to calculate simple interest using form.
35. Write a PHP program to demonstrate different file-handling operations using a text file. Create an HTML form through which the user can enter a filename and perform the following operations:
 - a. Check File: Check whether the specified file exists and display an appropriate message.
 - b. Open File: Open the file using `fopen()` in the required mode.
 - c. Write File: Accept text from the user and write it to the file using `fwrite()`.
 - d. Read File: Read and display the contents of the file using `fread()` or `fgets()`.
 - e. Close File: Close the file using `fclose()` after completing the required operation.
 - f. Rename File: Accept a new filename and rename the existing file using `rename()`.
 - g. Check Permissions: Display the current permissions of the file using `fileperms()`.
 - h. Change Permissions: Change the file permissions using `chmod()` and display the updated permissions.
36. Write a PHP program to calculate the annual income tax of an individual according to the Nepal Government's income tax rates for Fiscal Year 2083/84. Create an HTML form that accepts the taxpayer's annual taxable income and **gender** as input. And calculate the tax progressively according to the following tax slabs:
 - Up to NPR 1,000,000: 1% tax
 - Next NPR 500,000 (NPR 1,000,001–1,500,000): 10% tax
 - Next NPR 1,000,000 (NPR 1,500,001–2,500,000): 20% tax
 - Next NPR 1,500,000 (NPR 2,500,001–4,000,000): 27% tax
 - Income above NPR 4,000,000: 29% tax
 - For female taxpayer, apply 10% discount on total tax

After form submission, display the following information:

1. Annual taxable income

2. Tax calculated under each applicable slab
 3. Total tax payable
 4. Net income after tax
37. Design an HTML form and develop a PHP program to process and validate the submitted form data using server-side validation. The form should contain the following fields: Name, Address, Username, Email, Password, Website, Phone, Gender, and Course.

Implement the following validation requirements:

- **Name:** Must not be empty and must contain only alphabetic characters and spaces. It must not contain numbers or special characters.
- **Address:** Must not be empty.
- **Username:** Must not be empty and must contain only letters, numbers, and underscores (_). Spaces and other special characters are not allowed.
- **Email:** Must not be empty and must be in a valid email format.
- **Password:** Must not be empty and must contain at least 8 characters, including at least one uppercase letter, one lowercase letter, one digit, and one special character.
- **Website:** Validate that the entered value is a valid website URL.
- **Phone:** Must not be empty, must contain only digits, and must start with **96, 97, or 98**.
- **Gender:** The user must select one of the available gender options.
- **Course:** The user must select a valid course from the dropdown list.

Thank you

